

# Wenyu Chiou

## AI Agent Infrastructure & Evaluation Engineer

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I build AI-agent and research infrastructure — MCP pipelines, orchestration, and harnesses — and I study when LLM agents can be trusted to stand in for real people, using ground-truth data I collected myself.

Open to Summer 2027 internships in AI evaluation, research engineering, and risk analytics.

Work eligibility: F-1 student, CPT-eligible for internships (no sponsorship required).

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## — EXPERIENCE

**Graduate Research Assistant** — Complex Adaptive Water Systems Lab, Lehigh University Aug 2024 – present

- Lead an **LLM-agent validity** program — whether LLM agents reproduce real human decisions — tested against a **937-household survey I designed and fielded**; co-authored the published *Water Resources Research* (2026) study that substitutes LLMs as the behavioral engine of the simulation.
- Building **WAGF**, a governance framework that constrains LLM agents with domain rules and behavioral theory so coupled-simulation outcomes stay stable and physically sensible (in preparation for release).
- Built **FLOODABM**, a **52,141-household** coupled agent-based × catastrophe simulation with NFIP mechanics, calibrated with structural equation modeling and Bayesian methods, and validated against OpenFEMA insurance-claims data.

**Independent Open-Source AI Infrastructure** 2024 – present

- Built **research-hub**, an **MCP server + CLI** that turns Zotero, Obsidian, and NotebookLM into an AI-operable research workspace: searches scholarly databases, ranks results with BM25, and gates retrieval quality with **recall@k evaluation against golden fixtures**. **PyPI research-hub-pipeline v1.1.1**; listed in **Awesome MCP Servers**; CI across **3 OS × 4 Python versions**.
- Built a multi-agent tooling stack — **ai-research-skills** (15-skill research catalog), **fable-method-harness** (cost-routing + audit-trail harness), and **agent-collab-skills + codex-delegate** (orchestration + delegation) — on the **OpenAI and Anthropic APIs, MCP, and Ollama**, with anti-fabrication contracts verified from git state and CI on Ubuntu and Windows.
- **10 merged pull requests** into third-party open-source projects, including a root-caused engine-selection fix with a regression test that fails on main (BuilderIO/agent-native ×4) and a Windows file-permissions security fix (langchain-ai/openwiki).
- **awesome-agentic-ai-zh** — a trilingual agentic-AI learning curriculum with CI-enforced locale parity; **4.5k+ GitHub stars**, 588 forks.

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## — PUBLICATION

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111.

[doi.org/10.1029/2025WR042111](https://doi.org/10.1029/2025WR042111) Published

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## — EDUCATION

**Ph.D., Civil & Environmental Engineering**, Lehigh University 2024 – present · expected 2028

**M.S., Hydrological and Oceanic Sciences**, National Central University Sep 2021 – Jun 2023

**B.S., Earth Sciences**, National Central University Sep 2017 – Jun 2021

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## — SKILLS

**LLM & agent systems:** OpenAI & Anthropic APIs · Model Context Protocol (MCP) · multi-agent orchestration · agent governance & validation · retrieval (BM25) + recall@k evaluation · Ollama (local models)

**Research automation:** Zotero · Obsidian · NotebookLM · arXiv / Semantic Scholar / PubMed / CrossRef

**Programming & tooling:** Python · pytest & CI (GitHub Actions) · Git · cross-platform (Windows / POSIX)